Spatial Cognition or Pattern Matching ? Re examining Fine-Tuned LLMs on Urban Computing Benchmarks

Group 30
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1 Introduction and summary of the selected paper

Large language models (LLMs) are increasingly proposed as components of urban systems such as navigation assistants, planning support and mobility simulation. Whether they actually understand city space is usually argued from benchmark scores. This project aims to examine how far such scores can be trusted.

The selected paper. CityGPT [1] sets out to give LLMs city-scale spatial cognition. Its authors simulate human mobility on real maps (OpenStreetMap road networks with Foursquare POIs) and turn the simulated experience into an instruction-tuning dataset, *CityInstruction*. It contains single-entity questions (CityQA), multi-step routes (CityWalk) and step-by-step direction and distance reasoning (CityReasoning). They fine-tune 6–8B models (ChatGLM3, Qwen2.5, Llama3) on this data mixed with general instruction data, optionally with a self-weighted loss (SWFT). They evaluate on *CityEval*, a multiple-choice benchmark with three scored groups: City Image, Urban Semantics and Spatial Reasoning (SR). On SR, fine-tuned Qwen2.5-7B improves from 0.164 to 0.552 in Beijing (+237%) and beats models up to 405B parameters. Models trained on Beijing also improve in London, New York and Paris, which the authors read as transferable spatial knowledge. Code and data are public [2, 3], model checkpoints are not.

State of the art. Outside the urban domain, benchmarks are known to contain artefacts that models exploit instead of the intended skill (*shortcut learning* [4]). Such artefacts are classically exposed with baselines that see only part of the input [5, 6]. To my knowledge, no such audit exists for CityEval.

The new problem. My inspection of the released benchmark, training data and code (CityEval v2.3) shows that CityEval's SR scores can be inflated without any spatial knowledge:

- **The answer options give the answer away.** Distance distractors are generated from the true distance d as $[d/2, 2d, d{+}500]$, or $[d/2, 2d, d{-}1000]$ for POI-to-POI questions. So d is *always* the 2nd- (or 3rd-) smallest option. This rule scores 1.00 on every distance task in all four cities, and 9.5% of distance items even offer a distance ≤ 0.
- **This matters for the headline numbers.** Distance tasks are exactly half of the SR score, so a rule that never reads the question and guesses on direction is expected to score 0.625. That is at least as high as every SR score the paper reports (best: 0.619), assuming its tables used the released v2.3 tasks.
- **The training data teaches the rule.** All four CityInstruction files share 15,258 San Francisco items in the test format: 7,266 in the exact CityEval templates and 7,992 CityReasoning items. About 4,700 numeric multiple-choice items among them follow the same rank rule in 100% of cases. There is no overlap with the test questions themselves.
- **Part of the gain may be formatting.** Unparseable answers are scored as wrong, and several base models score *below chance*. Fine-tuning may partly just teach the expected answer format.
- **The paper's own results fit this pattern.** Llama3-8B trained on Beijing scores about 0.6 on *no-context* SR questions in cities it never saw (base: about 0.2). That is hard to explain by map knowledge.

The question is therefore whether fine-tuning on CityInstruction teaches city space or teaches the test set.

Novelty and research challenge. I identify a weakness of the paper and try to fix it. I measure, with controlled experiments, how much of the fine-tuning gain is spatial knowledge and how much is shortcut, template familiarity and answer formatting. I then fix the training data and test whether robust gains remain. I also compare fine-tuning with a classic GIS baseline and with retrieval. The research challenge is not engineering,

it is designing transformations of benchmark items that change their *form* while provably preserving their *spatial content*, metrics that stay comparable across formats, and controls that isolate one factor at a time, all for items that are not statistically independent.

2 Problem statement

Items and models. For cities $c \in \{\text{Beijing}, \text{London}, \text{New York}, \text{Paris}\}$, a CityEval item is $q = (x, O, y)$ with question text x, options $O = \{o_1, \ldots, o_k\}$ and correct answer $y \in O$. Items form task sets; D^{dis} and D^{dir} denote distance and direction items. Every SR item exists with and without route context. A model f maps (x, O) to an option, and

$$\text{Acc}(f, D) = \frac{1}{|D|} \sum_{(x,O,y)\in D} \mathbb{1}\left[f(x, O) = y\right]. \tag{1}$$

I compare a base model f_0 with models fine-tuned on Beijing (the paper's transfer setting), the standard model f_{FT}, a template ablation $f_{-\text{task}}$, and a fixed-data model f_{fair} (section 4).

Regularities that can inflate the gain.

- **R1, option shortcut:** the rule $g(O)$, "pick the 2nd-smallest option" (3rd for POI-to-POI), recovers y without reading x.
- **R2, template familiarity:** training items share the exact test templates.
- **R3, answer-format compliance:** fine-tuning teaches the output format the evaluation parser expects.

Quantities.

- **Fine-tuning advantage:** $A(D) = \text{Acc}(f_{\text{FT}}, D) - \text{Acc}(f_0, D)$.
- **Transformations.** A transformation T rewrites items while preserving their spatial content: fair options T_{opt}, removed options T_{open}, paraphrased questions T_{para}. Its effect is the *drop* $\delta_T = A(D) - A(T(D))$.
- **Rule-following rate.** On $D' = T_{\text{opt}}(D^{\text{dis}})$, wheree distractors are regenerated so that g is right only by chance $(1/k)$, shortcut use is measured as

$$\text{RF}(f) = \frac{1}{|D'|} \sum_{(x,O',y)\in D'} \mathbb{1}\left[f(x, O') = g(O')\right]. \tag{2}$$

  A model that uses spatial information has RF $\approx 1/k$. One that follows the rule keeps choosing g's option even when it is wrong. Because base models may prefer middle values anyway, shortcut learning is measured relative to the base model, as $\text{RF}(f_{\text{FT}}) - \text{RF}(f_0)$.

3 Research questions

Main question: *To what extent do the spatial reasoning gains from fine-tuning on CityInstruction reflect urban spatial cognition rather than regularities of the CityEval format, and can the training data be fixed so that the gains become robust?*

RQ1 (shortcut use). To what extent do models fine-tuned on CityInstruction answer CityEval distance questions by the option-rank rule rather than from spatial information, as measured by their rule-following rate when the rule is made uninformative and by their accuracy on knowledge free items?

RQ2 (format dependence). How much of the fine-tuning advantage over the base model remains when answer-format compliance is controlled by likelihood scoring, when options are made fair or removed, and when questions are paraphrased? Where the option shortcut cannot help, how much of the advantage is attributable to training on test-template items, at equal training-set size?

RQ3 (fix). Does fine-tuning on CityInstruction with shortcut-free distractors produce gains that persist on the shortcut-free benchmark, follow the rank rule no more than the base model does, and lose less of their advantage when the answer options are removed than those of standard fine-tuning?

RQ4 (classic vs. foundation model). On no-context questions whose places can be geocoded, how do a classic GIS baseline, a retrieval-augmented base LLM and a fine-tuned LLM compare, in the training city and in unseen cities?

Each RQ has one primary test, fixed before any model is trained (section 5). Each outcome is informative either way. For example, if RQ1 finds no rule-following, the benchmark flaw exists but does not drive model scores, which supports the paper's claims.

4 Methodology

Models and training data. No checkpoints are released, so I train my own: Qwen2.5-1.5B-Instruct [7], fine-tuned with LoRA [8] on the Beijing CityInstruction file. To save compute, I use a reduced training set (78,491 instead of 138,356 items; about 19M instead of 62M tokens). It keeps all city, reasoning, template and spatial items, but only 10,000 of the 30,000 CityWalk route items and 2,000 of the 41,865 general instruction items, which no RQ needs. I train four kinds of model, all with one fixed LoRA setup (rank 16, 1 epoch) that is never tuned on CityEval results:

- f_{FT}: standard fine-tuning (3 seeds).
- $f_{-\mathrm{task}}$: without the 7,266 items that copy CityEval's question templates. f_{FT} instead leaves out 7,266 random items, so both models see the same amount of data.
- f_{fair}: the same items as f_{FT}, but with the shortcut removed from every training distractor set.
- $f_{\mathrm{FT}}^{\mathrm{full}}$: one run on the full training set, to check that the reduced set does not change the conclusions.

Models trained with one seed get two more if a result is too close to call.

Test conditions. Every model is tested on the original benchmark and on transformed versions of it (table 1). A transformation changes how an item is presented, not the spatial question it asks. Two further sets of *knowledge-free* items cannot be solved with anything learned in Beijing: questions without route context from unseen cities, and questions about fictional places. If fine-tuning improves distance but not direction accuracy on these items, that points to the shortcut. Direction items have no option shortcut, so they act as a built-in control. For the same reason, the template ablation ($f_{-\mathrm{task}}$ against f_{FT}) is compared only where the shortcut cannot help: on direction items and on T_{opt} and T_{para} items.

Operator	What it does	Tests for	Validity check
T_{opt}	replaces the distractors with values drawn from the same task's real answers, so the rank rule stops working	R1	a classifier that sees only the options must score at chance
T_{open}	removes the options, the model's free answer is mapped to the nearest option	use of options	share of answers that cannot be mapped
T_{para}	rewords the question with an LLM from another model family, fair options for distance items	R2	names and numbers unchanged. 100 items checked by hand

Table 1: Transformations of benchmark items.

Scoring and sample. For all comparisons, a model's answer is the option letter to which it gives the highest likelihood (P_{ll}), so answer formatting (R3) cannot affect the score. The paper's own protocol, free generation plus its answer parser (P_{gen}), is used only to reproduce its results and to count unparseable answers. Each model is tested on one fixed sample of 100 items per task and city (8,000 SR items).

Course techniques. The project draws on several course topics:

- validating machine learning on synthetic urban data: CityEval is simulator-generated, and I systematically perturb it;
- spatial data and dependence: the benchmark's places are points, lines and polygons, and by Tobler's first law [9] items about the same places are not independent;
- blocked evaluation: testing in cities never seen in training is a spatially blocked split rather than a random one [10];
- foundation models and fine-tuning [11];
- trajectories (the route context is a compressed trajectory) and the visualisation guidelines for all figures.

Resources and feasibility.

- **Compute.** Free resources only: the LIACS REL Compute (DSLab) node `vibranium` (2× RTX 3090 with 24 GB each, 256 GB RAM, 3.4 TB local disk), and a laptop (RTX 3060, 6 GB) for development.
- **Training code.** The paper's bundled training framework is incomplete (its data module is missing, and the fork it references no longer exists). I therefore use the maintained upstream LLaMA-Factory, which supports all the options of the paper's own LoRA configuration. The paper's evaluation code is complete and is reused.

Risks and mitigation. Table 2 lists the main risks.

Risk	Mitigation
The 1.5B model does not reproduce the paper's gain	checked after the first training run; if it fails, switch to Qwen2.5-3B
The reduced training set changes what is learned	compare with the full-data run $f_{\mathrm{FT}}^{\mathrm{full}}$; if they disagree, retrain $f_{-\mathrm{task}}$ and f_{fair} on the full set (+14 GPU-hours)
My models are smaller than the paper's and use LoRA	the reproduction check; the paper's own transfer results as independent evidence; conclusions stated for my setting
`vibranium` is busy (shared, no job queue)	overnight runs; the student node `tritanium` for evaluation; development on the laptop
A transformation changes an item's meaning	the validity checks in table 1
Items are not independent	cluster-based statistics (section 5)

Table 2: Risks and mitigation plans.

5 Evaluation approach

Metrics. The main metrics are accuracy (averaged over tasks, as in the paper), the fine-tuning advantage A and how much of it each transformation removes (δ_T), the rule-following rate RF, and accuracy on the knowledge-free items. As a further check, I relate accuracy to the gap between the correct answer and the nearest wrong option: a model that uses real knowledge should become more accurate as the gap grows, whereas the rule is unaffected by it. Each RQ has one primary measure, fixed in advance; all other analyses are secondary:

- **RQ1:** $\mathrm{RF}(f_{\mathrm{FT}}) - \mathrm{RF}(f_0)$;
- **RQ2:** $\delta_{T_{\mathrm{opt}}}$;
- **RQ3:** the advantage of f_{fair} on $T_{\mathrm{opt}}(D)$, together with its rule-following and option-removal criteria;
- **RQ4:** the retrieval-augmented base model f_{RAG} against f_{FT}.

Baselines. The base model f_0; the rule g, which never reads the question; random guessing; the paper's reported numbers; and, as an upper reference for with-route items, a parser that solves them directly from the route text. For RQ4, a classic GIS baseline (compass bearing, and straight-line distance scaled by a road-detour factor estimated in Beijing) and f_{RAG}.

Statistics. Items are not independent: the same routes produce items for all ten SR task types, and 58–93% of sampled items share a place with another item. I therefore group items by their starting location (at most 5–10 items per group) and always resample whole groups. Confidence intervals come from a bootstrap over seeds and groups, and p-values from group-level permutation tests (with 1 km spatial blocks for RQ4) Results are reported separately for Beijing and for the unseen cities. With 100 items per task, differences of about 7 points per city and condition, or about 2 points pooled, are detectable.

6 Data sources and other resources

All sources were checked to exist and to be accessible on 31 October 2026. All are free; nothing is paid.

- **CityInstruction and CityEval v2.3** [3]: https://huggingface.co/datasets/Tianhui-Liu/CityGPT-Data (MIT licence; 1.4 GB; already downloaded and audited).
- **CityGPT code** [2]: https://github.com/tsinghua-fib-lab/CityGPT (MIT licence). Model checkpoints are not released anywhere, which is why I train my own.
- **Paper** [1]: https://doi.org/10.1145/3711896.3736878, preprint at https://arxiv.org/abs/2406.13948.
- **Models**: Qwen2.5-0.5B/1.5B-Instruct [7] from Hugging Face (open weights).
- **Training framework**: LLaMA-Factory (https://github.com/hiyouga/LlamaFactory); vLLM for generation.

Ethical statement

The data consists of simulator-generated questions about public map data (OpenStreetMap roads and Foursquare places) under a permissive licence. It contains no personal data. CityGPT's composite mobility tasks, which

use real user trajectories, are outside this project's scope. This audit is only possible because the authors released their code and data openly. The societal relevance is that benchmark scores guide whether LLMs are trusted in urban uses such as navigation or planning, so it matters what those scores measure. My results may well support the paper's claims. If they do not, they could be misread as evidence that LLMs have no spatial knowledge at all, so I report what survives the controls as well as what does not. The benchmark covers four large, well-documented cities, so conclusions may not transfer to cities that are underrepresented in web data.

Code

The project repository is https://github.com/Rahulrayy/UC-Project. It will contain the data pipeline, the benchmark transformations, the evaluation and statistics code, and the analysis scripts, and it will be updated throughout the project.

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